

IZMA AZIZ

SENIOR SOFTWARE ENGINEER – GENERATIVE AI & AGENTIC SYSTEMS

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SUMMARY

Senior AI Engineer with 4+ years of experience in Generative AI, LLMs, and Multi-Agent Systems. Led AI delivery for a self-hosted SaaS chatbot platform with one-click deployment, enabling isolated customer environments. Improved response accuracy by up to 80% through optimization of retrieval, model performance, and system reliability. Microsoft Certified (4 Azure specializations). Bronze Medalist, NED University. Speaker and blogger on Generative and Agentic AI.

TECHNICAL SKILLS

- **AI / LLM Systems:** Generative AI, LLM Applications, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Agentic Workflows, Prompt & Context Engineering, ReAct Agents, Model Context Protocol (MCP), Text-to-SQL
- **LLM Frameworks, Retrieval & Observability:** LangChain, LangGraph, LlamaIndex, Langfuse, Vector Databases, FAISS, ChromaDB, Hybrid Search, Embeddings, Re-ranking
- **Backend, Data & APIs:** Python, SQL, Django, FastAPI, REST APIs, Async Processing, PostgreSQL, Cosmos DB, ADLS Gen2, ETL Pipelines
- **Cloud & Azure AI Platforms:** Azure OpenAI, Azure AI Foundry, Azure AI Search, Azure Functions, Azure Web Apps, Azure Synapse, Azure Document Intelligence
- **DevOps & LLMOps:** Docker, Git, GitHub Actions, Azure DevOps, ARM Templates, Azure Monitor, LLM Observability
- **System Design & Architecture:** Microservices, Event-Driven Systems, Scalable AI Systems, High Availability, Fault Tolerance, API Design
- **Leadership & Collaboration:** Technical Leadership, Cross-team Influence, Client Engagement, Scrum Facilitation, Mentorship, Technical Interviewing, Documentation & Design Ownership

WORK EXPERIENCE

Senior Software Engineer - Generative AI and LLMs

Aug 2025 — Present

Data Science Dojo LLC | Remote, United States

- **Owned end-to-end architecture and delivery of agentic AI systems** for a multi-tenant enterprise platform **serving 10+ clients**, leading a **team of 7–8 engineers**, conducting **30+ technical interviews**, and mentoring engineers to improve team capability and delivery standards.
- **Defined and implemented deep-search and multi-agent reasoning architectures** (LangGraph), introducing hierarchical planning (supervisor–sub-agent orchestration) that improved task success and retrieval accuracy by **70%+ in production**.
- **Led context engineering strategy** by designing hybrid **memory systems** (short-term conversational + long-term semantic with user/thread isolation), improving response relevance and personalization by **50%**.
- **Established agent execution frameworks** enabling dynamic tool use, prompt composition, and runtime capability binding, improving decision efficiency and reducing unnecessary LLM calls.
- **Drove system reliability and cost optimization initiatives**, implementing structured routing, retry-safe execution, and recovery mechanisms, achieving **90% uptime** and reducing compute **costs by 40%**.
- **Owned observability, evaluation, and governance of AI systems**, implementing tracing (Langfuse), human-in-the-loop workflows, and safety guardrails to ensure controlled, auditable, and production-safe AI behavior.

Data Scientist I & II

Dec 2023 — Aug 2025

Data Science Dojo LLC | Remote, United States

- **Built and optimized production-grade RAG systems**, improving response relevance, reducing hallucinations, and achieving up to 80% client satisfaction across enterprise deployments.
- **Designed and scaled enterprise retrieval pipelines** using hybrid search (semantic, keyword, and query-time augmentation), improving coverage and answer quality across structured and unstructured data sources.
- **Developed graph-based, user-configurable AI workflow systems**, enabling dynamic pipeline construction through modular components (agents, tools, retrieval, prompts).
- **Implemented ReAct-based agents with self-critique mechanisms**, improving reasoning accuracy and enabling adaptive task execution across use cases.
- **Established evaluation and monitoring frameworks (RAGAS)** that enable systematic measurement of retrieval accuracy, faithfulness, and response quality for continuous system improvement.

Data Science Intern & Trainee

Jun 2023 — Dec 2023

Data Science Dojo LLC | Remote, United States

- Built **workflow automation pipelines using Azure Logic Apps**, integrating data ingestion, processing, and storage systems for scalable execution.
- Worked on a **client-facing RAG system**, applying **prompt engineering** and response optimisation techniques to improve relevance, **reduce hallucinations by 60%**, and **achieve 80% client satisfaction**.

Machine Learning Intern

Jun 2022 — Jul 2022

ConceptZI | Remote, Pakistan

- Worked on **data preprocessing and feature engineering** for machine learning pipelines
- Applied foundational **machine learning and deep learning techniques** on structured datasets

IT Intern

Apr 2022 — Jun 2022

Muller & Phipps Pakistan (Private) Limited | Karachi, Pakistan

- Performed **quality assurance testing** across internal systems and software workflows
- Supported **data reporting and analysis using Power BI**, including data cleaning and visualization

Web Development Intern

Oct 2021 — Jan 2022

VIRTUOSOFT (NIC NEDUET) | Karachi, Pakistan

- Built foundational understanding of backend development using **Django framework and REST-based architecture**

EDUCATION

Bachelor's in Computer and Information Systems Engineering

Oct 2019- Aug 2023

NED University of Engineering and Technology Karachi, Pakistan

- **CGPA: 3.962 (Bronze Medalist)**
- **Thesis: "Smart SQL Query Generation" (Group Leader)** - Fine-tuned T5 transformer for NL-to-SQL conversion, achieving 80% ROUGE precision on Spider benchmark (39% above baseline)

CERTIFICATIONS

- [Microsoft Certified: Azure Solutions Architect Expert \(AZ-305\)](#)
 - [Microsoft Certified: Azure AI Engineer Associate \(AI-102\)](#)
 - [Microsoft Certified: Azure Administrator Associate \(AZ-104\)](#)
 - [Anthropic: Building with the Claude API](#)
 - [DeepLearning.AI: Neural Networks and Deep Learning](#)
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AWARDS

- **Enabler Award - Data Science Dojo:** Contribution to team capability, mentorship, and high-performance delivery
 - **DOT Awards (4x) & Giant Leap Award - Data Science Dojo:** Excellence in project delivery and impactful AI/ML contributions
 - **Bronze Medal, B.E. Computer Systems Engineering - NED University of Engineering and Technology:** Ranked 3rd out of 152 graduating students
 - **Runner-Up, Developers Day 2022 Data Science Competition - National University of Computer & Emerging Sciences (FAST-NUCES)**
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PROJECTS

- [Multi-Agent Deep Agent with Skills Demo](#)

Tech: LangChain, Azure OpenAI, Streamlit

Built a planning-oriented deep agent with a custom skill framework using a progressive disclosure strategy, enabling dynamic multi-agent coordination

- [Agentic AI System using LlamaIndex](#)

Tech: LlamaIndex, Azure OpenAI

Implemented a ReAct agent using LlamaIndex event-driven workflows with dynamic tool selection and self-reflection for adaptive agent behavior

- [RAG-Based Conversational AI System](#)

Tech: LangChain, ChromaDB, LLMs

Built a retrieval-augmented chatbot with efficient prompt structures and retrieval strategies tuned for scalable, production-ready performance across large enterprise knowledge bases

- [Smart SQL Query Generation \(Thesis\)](#)

Tech: Python, T5 Transformer, NLP, Spider & WikiSQL

Fine-tuned T5 transformer for NL-to-SQL conversion, achieved 80% ROUGE precision on Spider benchmark through schema-aware input formatting, synonym augmentation, and hyperparameter tuning, 39% above baseline, outperforming BERT and SQLNet

PUBLICATIONS

Webinar - Data Science Dojo:

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|---|----------|
| • Scaling AI Beyond Single Agents: Multi-Agent Architectures with LangChain | Feb 2026 |
| • Master Agentic AI: Event-Driven LlamaIndex Workflows | Feb 2025 |
| • Building Simple and Efficient Chatbots Step-by-Step | Feb 2024 |

Blogs on Generative AI - Data Science Dojo:

- Authored blog posts on Dynamic Few-Shot Prompting, RAG in LLM, and Evolution of the GPT Series.